

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-12	Team No-02	Project Title-Vehicle Service Booking & Tracking System
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1. Abstract

- The **Vehicle Service Booking and Tracking System** is a web-based application developed to simplify the process of booking, managing, and tracking vehicle maintenance services. The system allows customers to schedule service appointments online by selecting the required service from a predefined catalog. It eliminates the traditional manual booking process and provides a convenient platform for both customers and service center administrators. The application also enables customers to view their service history, receive reminders for upcoming maintenance, and access digital invoices.
- The system incorporates a Job Card and Technician Allocation module that automatically generates service job cards and assigns technicians based on their availability and specialization. A Live Status Tracking feature keeps customers informed about the progress of their vehicles through real-time notifications. The system also maintains a Spare Parts Inventory to monitor the availability of vehicle components required during servicing. This reduces delays and improves service efficiency.
- The application is developed using React.js for the front-end user interface, Node.js and Express.js for server-side processing, and MySQL for database management. Firebase Cloud Messaging (FCM) is integrated to provide instant status updates and notifications, while Razorpay is used to support secure online payments. The proposed system improves communication between customers and service centers, enhances operational efficiency and user-friendly vehicle service management solution.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090205	A.Jhansi Priya	Jhansi
2520090177	A.Keerthi Priya	Keerthi

3. Frameworks / Platforms / Software Used

- **React.js** – Front-end framework for developing the user interface.
- **Node.js** – Runtime environment for server-side development.
- **Express.js** – Framework for building APIs and handling requests.
- **MySQL** – Database for storing application data.
- **Firebase Cloud Messaging (FCM)** – Platform for sending notifications and status updates.
- **Razorpay** – Payment gateway for online transactions.
- **Visual Studio Code** – Software used for coding and debugging.
- **Postman** – Tool used for API testing.
- **Git** – Version control system for tracking code changes.
- **GitHub** – Platform for managing and storing source code.

Guide Name and Signature (remarks if any) __